FET small-signal I

Practice questions · Day 18 · Tue 10 Nov · pack E11

5. Practice

From your course files

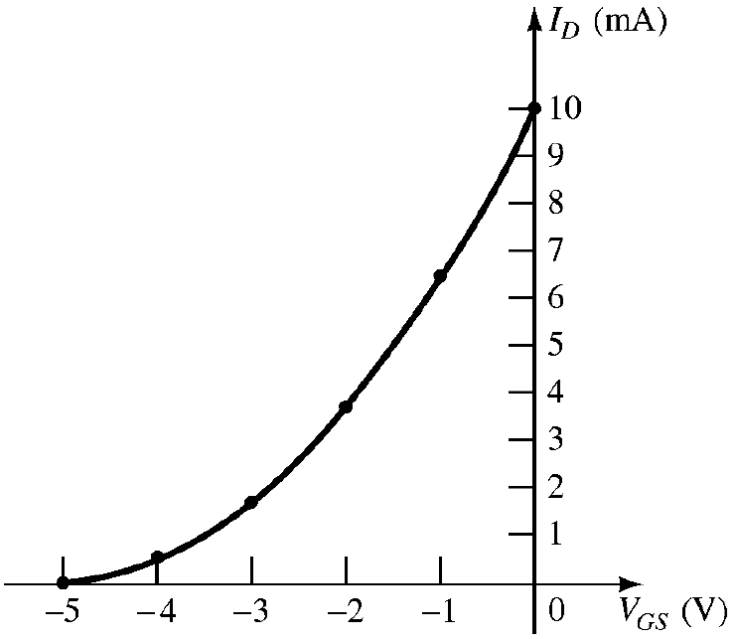
WS8 Q7 (course question). Calculate g_{m0} for a JFET having device parameters $I_{DSS} = 12$ mA and $V_P = -4$ V.

WS8 Q8 (course question). For a JFET having device parameters $g_{m0} = 5$ mS and $V_P = -4$ V, what is the device current at $V_{GS} = 0$ V?

WS8 Q9 (course question). For a JFET having specified values of $g_{fs} = 4.5$ mS and $g_{os} = 25$ μS, determine the device output impedance Z_o (FET) and device ideal voltage gain A_v (FET).

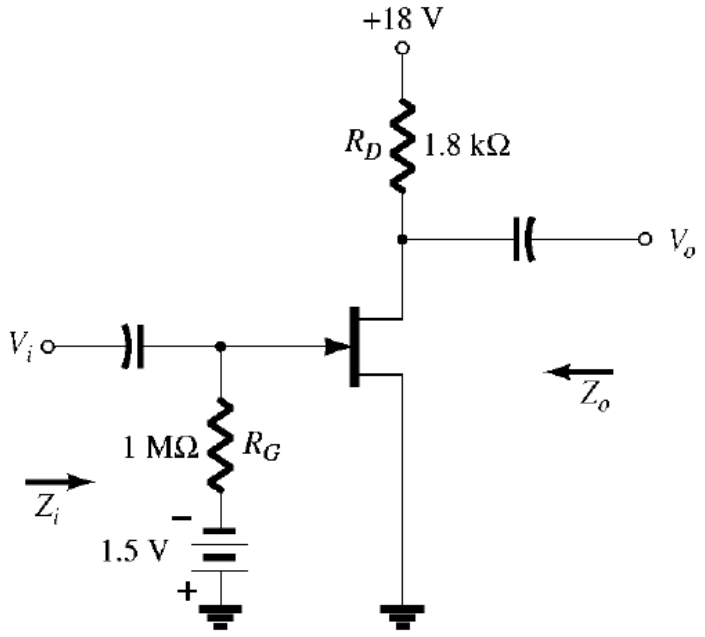
WS8 Q10 (course question). Using the transfer characteristic of Fig. 7:

a. What is the value of g_{m0}?
b. Determine g_m at $V_{GS} = -0.5$ V graphically.
c. Calculate the value of g_m at $V_{GS} = -0.5$ V and compare with the solution to (b).



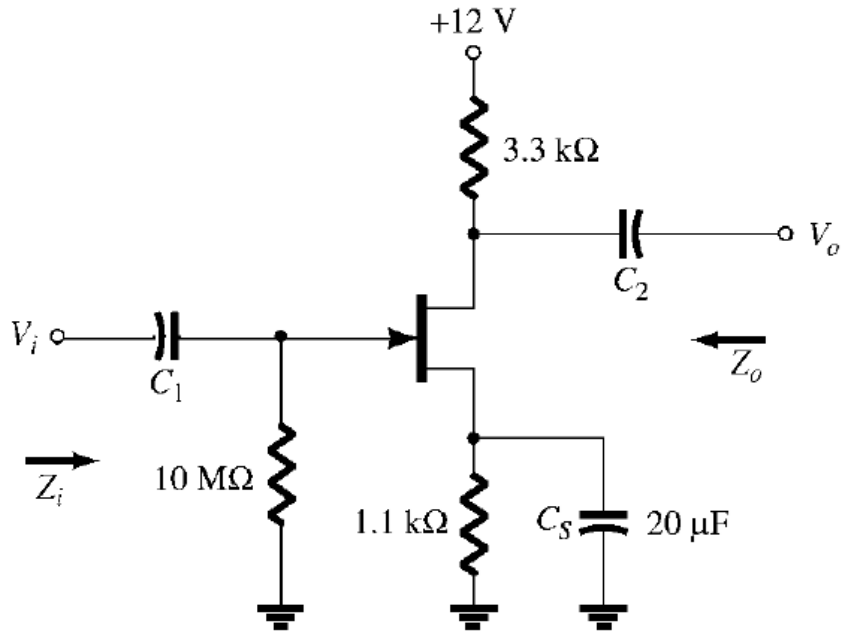
WS8 Fig. 7 (from the workshop sheet)

WS8 Q11 (course question). Determine Z_i, Z_o, and A_v for the network of Fig. 8 if $I_{DSS} = 10$ mA, $V_P = -6$ V, and $r_d = 40$ kΩ.



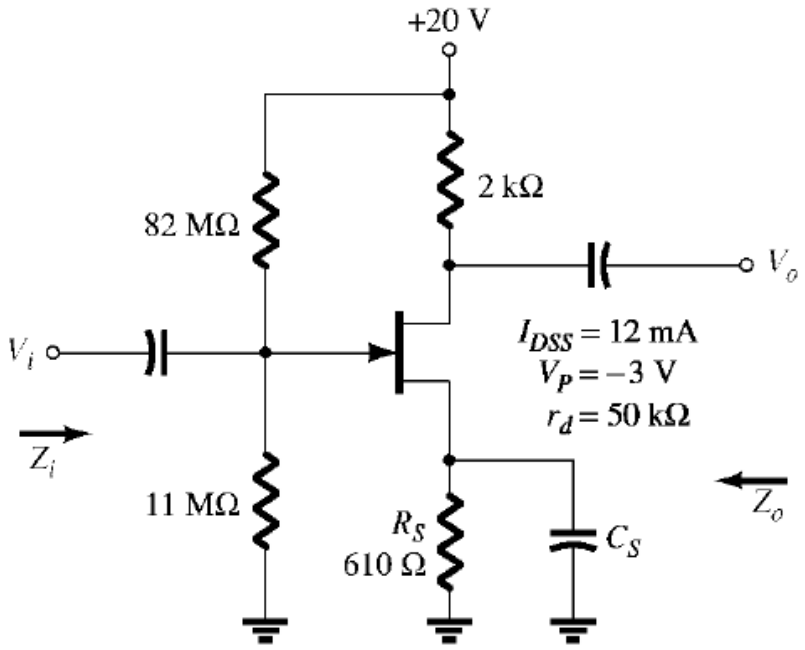
WS8 Fig. 8 (from the workshop sheet)

WS8 Q12 (course question). Determine Z_i and Z_o for the network of Fig. 9. Given: $g_{fs} = 3000\ \mu$S and $g_{os} = 50\ \mu$S.



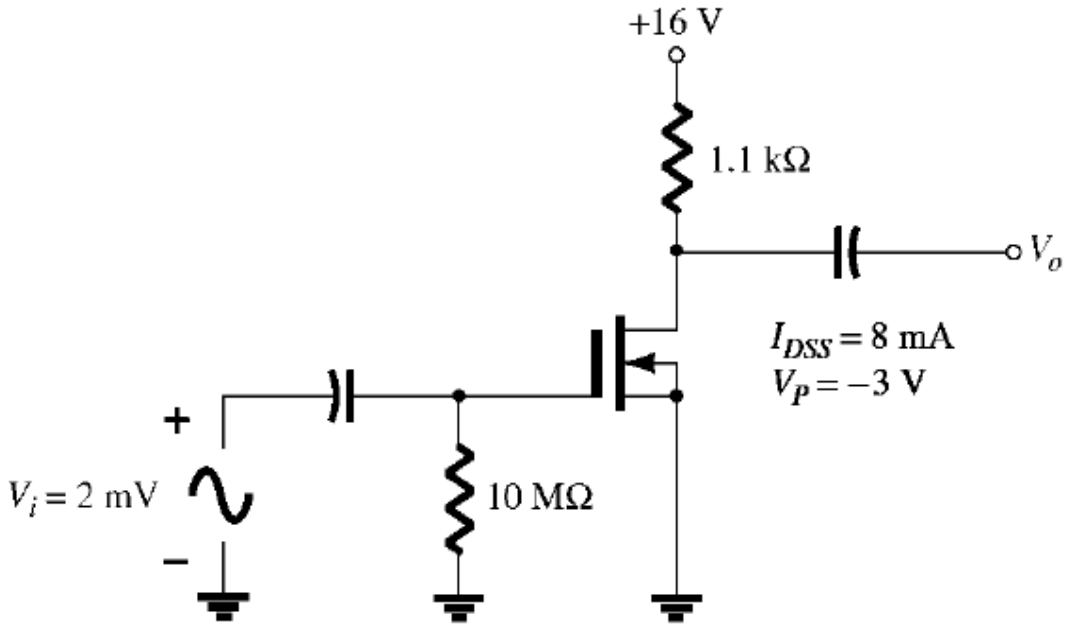
WS8 Fig. 9 (from the workshop sheet)

WS8 Q13 (course question). Determine Z_i, Z_o, and V_o for the network of Fig. 10 if $V_i = 20$ mV.



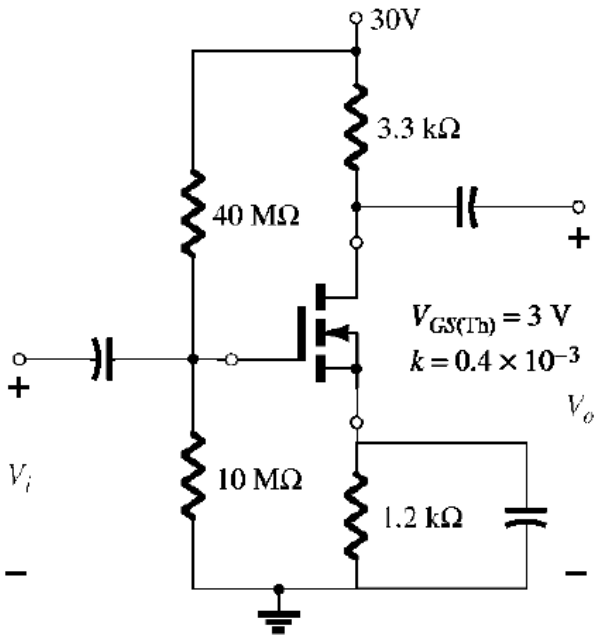
WS8 Fig. 10 (from the workshop sheet)

WS8 Q14 (course question). Determine V_o for the network of Fig. 11 if $g_{os} = 20\ \mu$S.



WS8 Fig. 11 (from the workshop sheet)

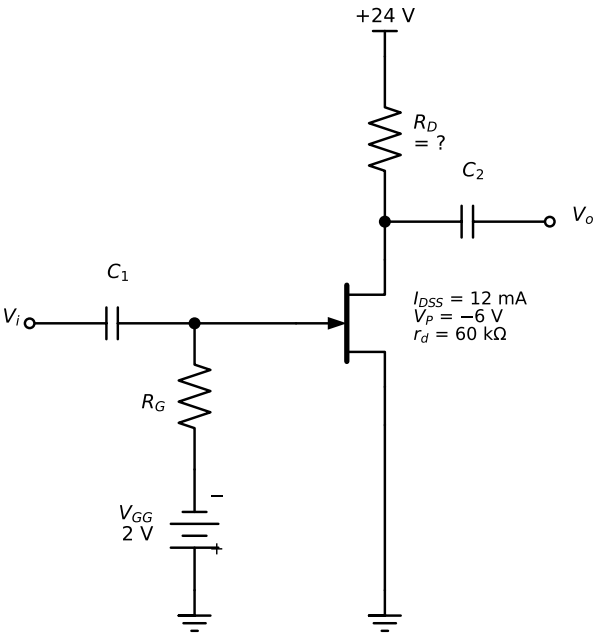
WS8 Q15 (course question). Determine the output voltage for the network of Fig. 12 if $V_i = 0.8$ mV and $r_d = 40$ kΩ.



WS8 Fig. 12 (from the workshop sheet)

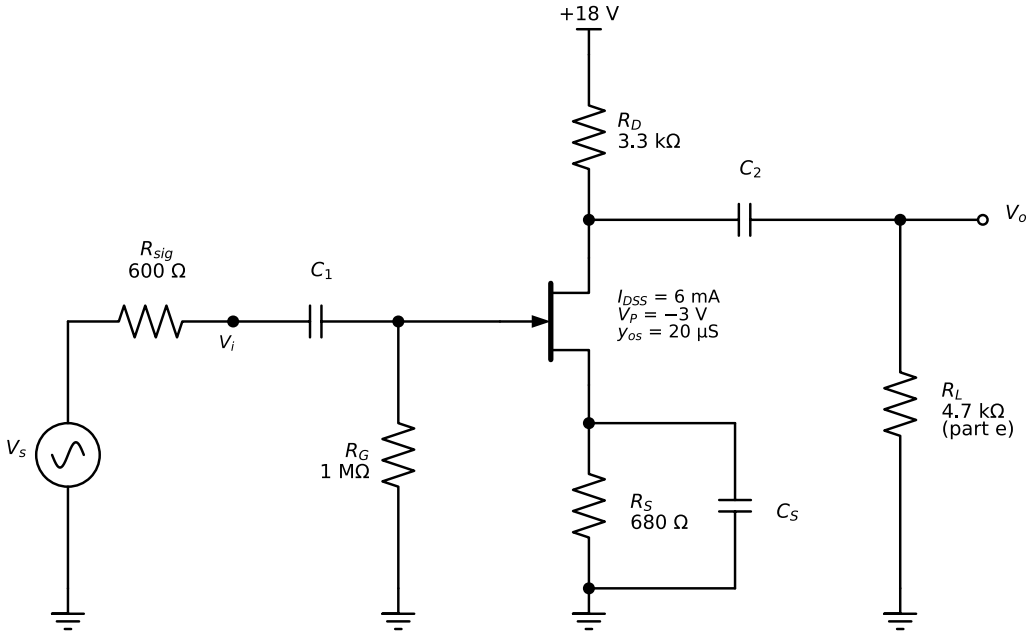
New questions

N1 [3 marks] (easy). A JFET has $I_{DSS} = 8$ mA and $V_P = -4$ V. (a) Find g_{m0}. (b) Find g_m at $V_{GS_Q} = -1.5$ V. (c) Find g_m at $I_{D_Q} = 2$ mA without first finding V_{GS}.



Circuit for N2: fixed-bias JFET amplifier, R_D to be chosen

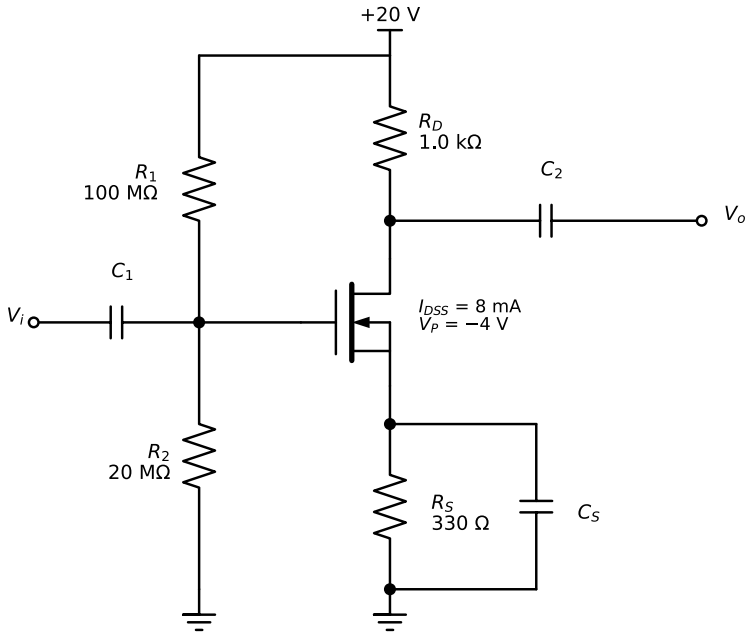
N2 [4 marks] (design). A fixed-bias JFET amplifier has $V_{DD} = 24$ V and gate bias $V_{GG} = 2$ V (so $V_{GS_Q} = -2$ V). The device has $I_{DSS} = 12$ mA, $V_P = -6$ V and $r_d = 60$ kΩ. (a) Find R_D for $A_v = -6$. (b) Find V_{DS_Q} with that R_D and confirm saturation.



Circuit for N3: self-bias JFET stage with source resistance and load

N3 [8 marks] (multi-part, chained). A self-bias JFET stage has $V_{DD} = 18$ V, $R_G = 1$ MΩ, $R_D = 3.3$ kΩ, $R_S = 680\ \Omega$. The device has $I_{DSS} = 6$ mA, $V_P = -3$ V, $y_{os} = 20$ µS.

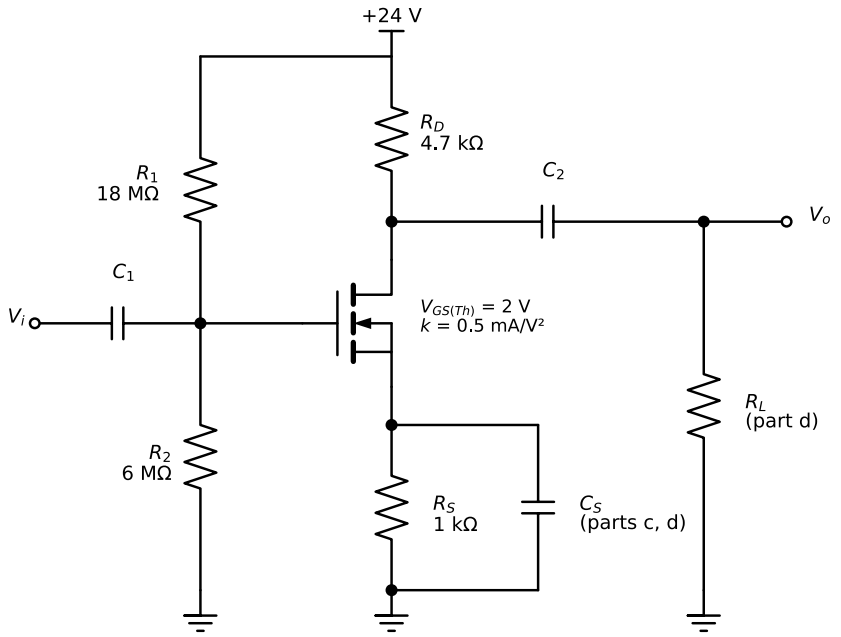
a. Find V_{GS_Q}, I_{D_Q} and V_{DS_Q}. [3]
b. Find g_m and r_d. [1]
c. With R_S fully bypassed, find Z_i, Z_o and A_v. [2]
d. The bypass capacitor fails open. Find A_v, ignoring r_d. [1]
e. With C_S working, a 4.7 kΩ load is connected and the source has $R_{sig} = 600\ \Omega$. Find $A_{vs} = V_o/V_s$. [1]



Circuit for N4: D-MOSFET voltage-divider stage

N4 [5 marks]. An n-channel D-MOSFET has $I_{DSS} = 8$ mA and $V_P = -4$ V, with r_d negligible. It is in a voltage-divider stage: $V_{DD} = 20$ V, $R_1 = 100$ MΩ, $R_2 = 20$ MΩ, $R_S = 330\ \Omega$ (bypassed), $R_D = 1.0$ kΩ.

a. Find the Q-point and check saturation.
b. Find g_m and compare it to g_{m0}.
c. Find Z_i and A_v.



Circuit for N5: E-MOSFET voltage-divider stage

N5 [6 marks] (hard). An n-channel E-MOSFET has $V_{GS(Th)} = 2$ V and $k = 0.5$ mA/V², with r_d ignored. It is in a voltage-divider stage: $V_{DD} = 24$ V, $R_1 = 18$ MΩ, $R_2 = 6$ MΩ, $R_S = 1$ kΩ, $R_D = 4.7$ kΩ.

a. Find V_{GS_Q}, I_{D_Q} and V_{DS_Q}. [2]
b. Find g_m. [1]
c. Find A_v with R_S unbypassed and with R_S bypassed. [2]
d. With R_S bypassed, what load R_L reduces $|A_v|$ to 5? [1]

6. Quick recall check

1. Write g_{m0} and g_m for a JFET. Which quantity takes an absolute value?
2. What fraction of g_{m0} is g_m when $I_D = I_{DSS}/4$?
3. Give Z_i, Z_o and A_v for a voltage-divider JFET with R_S bypassed.
4. Give A_v for self bias with R_S unbypassed, ignoring r_d.
5. Why can g_m exceed g_{m0} for a D-MOSFET but not for a JFET?